

Reorganizing the Financial Statements

Traditional financial statements—the income statement, balance sheet, and statement of cash flows—do not provide easy insights into operating performance and value. They simply aren’t organized that way. The balance sheet mixes together operating assets, nonoperating assets, and sources of financing. The income statement similarly combines operating profits, interest expense, and other nonoperating items.

To prepare the financial statements for analyzing economic performance, you should reorganize each financial statement into three categories: operating items, nonoperating items, and sources of financing. This often requires searching through the notes to separate accounts that aggregate operating and nonoperating items. This task may seem mundane, but it is crucial for avoiding the common traps of double-counting, omitting cash flows, and hiding leverage that distorts performance metrics, such as return on equity and cash flow from operations.

Since reorganizing the financial statements is complex, this chapter breaks down the process into three sections. The first section presents a simple example demonstrating how to build invested capital, net operating profit after taxes (NOPAT), and free cash flow. The second section applies this method to the financial statements for Costco Wholesale, with comments on some of the intricacies of implementation. Finally, we provide a brief summary of advanced analytical topics, including how to adjust for restructuring charges, operating leases, pensions, and capitalized expenses. An in-depth analysis of each of these topics can be found in the chapters of Part Three.

REORGANIZING THE ACCOUNTING STATEMENTS: KEY CONCEPTS

To calculate return on invested capital (ROIC) and free cash flow (FCF), it is necessary to reorganize the balance sheet to estimate invested capital, as well as to

likewise reorganize the income statement to estimate NOPAT. Invested capital represents the investor capital required to fund operations, without regard to how the capital is financed. NOPAT represents the after-tax operating profit (generated by the company's invested capital) that is available to all investors.

ROIC and FCF are both derived from NOPAT and invested capital. ROIC is defined as

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested Capital}}$$

and free cash flow is defined as

$$\text{FCF} = \text{NOPAT} + \text{Noncash Operating Expenses} - \text{Investment in Invested Capital}$$

By combining noncash operating expenses, such as depreciation, with investment in invested capital, it is also possible to express FCF as

$$\text{FCF} = \text{NOPAT} - \text{Increase in Invested Capital}$$

Invested Capital: Key Concepts

To build an economic balance sheet that separates a company's operating assets from its nonoperating assets and financial structure, we start with the traditional balance sheet. The accounting balance sheet is bound by the most fundamental rule of accounting:

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

The traditional balance sheet equation, however, mixes operating liabilities and sources of financing on the right side of the equation.

Assume a company has only operating assets (OA), such as accounts receivable, inventory, and property, plant, and equipment (PP&E); operating liabilities (OL), such as accounts payable and accrued salaries; interest-bearing debt (D); and equity (E). Using this more explicit breakdown of assets, liabilities, and equity leads to an expanded version of the balance sheet relationship:

$$\text{OA} = \text{OL} + \text{D} + \text{E}$$

Moving operating liabilities to the left side of the equation leads to invested capital:

$$\text{OA} - \text{OL} = \text{Invested Capital} = \text{D} + \text{E}$$

This new equation rearranges the balance sheet to reflect more accurately capital used for operations and the financing provided by investors to fund those operations. Note how invested capital can be calculated using either the operating method (that is, operating assets minus operating liabilities) or the financing method (debt plus equity).

For many companies, the previous equation is too simple. Assets consist of not only operating assets, but also nonoperating assets (NOA), such as marketable securities, prepaid pension assets, nonconsolidated subsidiaries, and other long-term investments. Liabilities consist of not only operating liabilities and interest-bearing debt, but also debt equivalents (DE), such as unfunded retirement liabilities, and equity equivalents (EE), such as deferred taxes and income-smoothing provisions. (We explain debt and equity equivalents in detail later in the chapter.) We can expand our original balance sheet equation to show these:

$$\begin{array}{ccccccccc} \text{OA} & + & \text{NOA} & = & \text{OL} & + & \text{D + DE} & + & \text{E + EE} \\ \text{(operating} & & \text{(nonoperating} & & \text{(operating} & & \text{(debt and its} & & \text{(equity and its} \\ \text{assets)} & & \text{assets)} & & \text{liabilities)} & & \text{equivalents)} & & \text{equivalents)} \end{array}$$

Rearranging leads to total funds invested:

$$\begin{array}{ccccccccc} \text{OA} - \text{OL} & + & \text{NOA} & = & \text{Total Funds} & = & \text{D + DE} & + & \text{E + EE} \\ \text{(invested} & & \text{(nonoperating} & & \text{Invested} & & \text{(debt and its} & & \text{(equity and its} \\ \text{capital)} & & \text{assets)} & & & & \text{equivalents)} & & \text{equivalents)} \end{array}$$

For a company with debt and equity equivalents, invested capital no longer equals debt plus equity. It equals operating assets minus operating liabilities. From an investing perspective, total funds invested equals invested capital plus nonoperating assets. From the financing perspective, total funds invested equals debt and its equivalents plus equity and its equivalents. Exhibit 11.1

EXHIBIT 11.1 An Example of Invested Capital

\$ million					
Accountant's balance sheet			Invested capital		
Assets	Prior year	Current year		Prior year	Current year
Cash	5	15	Cash	5	15
Inventory	200	225	Inventory	200	225
Net PP&E	300	350	Accounts payable	(125)	(150)
Equity investments	15	25	Operating working capital	80	90
Total assets	520	615			
			Net PP&E	300	350
Liabilities and equity			Invested capital	380	440
Accounts payable	125	150			
Interest-bearing debt	225	200	Equity investments	15	25
Shareholders' equity	170	265	Total funds invested	395	465
Total liabilities and equity	520	615			
			Reconciliation of total funds invested		
			Interest-bearing debt	225	200
			Shareholders' equity	170	265
			Total funds invested	395	465

Operating liabilities are netted against operating assets

Nonoperating assets are not included in invested capital

rearranges the balance sheet into invested capital for a simple hypothetical company with only a few line items. The reconciliation at the lower right shows how the amount of total funds invested is identical regardless of the method used.

Net Operating Profit after Taxes: Key Concepts

NOPAT is the after-tax profit generated from core operations, excluding any income from nonoperating assets or financing expenses, such as interest. Whereas net income is the profit available to equity holders only, NOPAT is the profit available to *all* investors, including providers of debt, equity, and any other types of investor financing. It is critical to define NOPAT consistently with your definition of invested capital and to include only those profits generated by invested capital.

To calculate NOPAT, we reorganize the accounting income statement in three ways (see Exhibit 11.2). First, interest is not subtracted from operating income, because interest is compensation for the company's debt investors, not an operating expense. By reclassifying interest as a financing item, we make NOPAT independent of the company's capital structure.

Second, when calculating NOPAT, exclude income generated from assets that were excluded from invested capital. Mistakenly including nonoperating income in NOPAT without including the associated assets in invested capital

EXHIBIT 11.2 An Example of NOPAT

\$ million			
Accountant's income statement		NOPAT	
	Current year		Current year
Revenues	1,000	Revenues	1,000
Operating costs	(700)	Operating costs	(700)
Depreciation	(20)	Depreciation	(20)
Operating profit	280	EBITA	280
Interest expense	(20)	Operating taxes ¹	(70)
Income from equity investments	4	NOPAT	210
Earnings before taxes (EBT)	264		
		Income from equity investments	4
Income taxes	(66)	Tax shield on nonoperating items ²	4
Net income	198	Income available to investors	218
		Reconciliation with net income	
		Net income	198
		Interest expense	20
		Income available to investors	218

Taxes are calculated on operating profits

Do not include income from any asset excluded from invested capital as part of NOPAT

Treat interest as a financial payout to investors, not an operating expense

¹ Assumes a marginal tax of 25% on all income.

² Interest tax shield less taxes on equity income.

will lead to an inconsistent definition of ROIC; the numerator and denominator will include unrelated elements. If one-time items such as a major litigation settlement are reported, exclude them from NOPAT as well. One-time items are important to analyze, but make trends in core performance difficult to identify.

Finally, since reported taxes are calculated after interest and nonoperating income, they are a function of nonoperating items and capital structure. Keeping NOPAT focused solely on ongoing operations requires that the effects of interest expense and nonoperating income also be removed from taxes. To calculate operating taxes, start with reported taxes, add back the tax shield from interest expense, and remove the taxes paid on nonoperating income. The resulting operating taxes should equal the hypothetical taxes that would be paid by an all-equity, pure operating company. Nonoperating taxes, the difference between operating taxes and reported taxes, are not included in NOPAT, but instead as part of income available to investors.

Free Cash Flow: Key Concepts

To value a company's operations, we discount projected free cash flow at a company's weighted average cost of capital. Free cash flow is the after-tax cash flow available to all investors: debt holders and equity holders. Unlike "cash flow from operations" reported in a company's annual report, free cash flow is independent of financing flows and nonoperating items. It can be thought of as the after-tax cash flow that would be generated if the company held only core operating assets and financed the business entirely with equity. Free cash flow is defined as:

$$\text{FCF} = \text{NOPAT} + \text{Noncash Operating Expenses} - \text{Investments in Invested Capital}$$

As shown in Exhibit 11.3, free cash flow excludes nonoperating flows and items related to capital structure. Unlike the accounting cash flow statement, the free cash flow statement starts with NOPAT (instead of net income). As discussed earlier, NOPAT excludes nonoperating income and interest expense. Instead, interest is classified as a financing cash flow.

Changes in nonoperating assets and the gains, losses, and income associated with these nonoperating assets are not included in free cash flow. Instead, nonoperating cash flows should be analyzed and valued separately. Combining free cash flow and nonoperating cash flow leads to cash flow available to investors. As is true with total funds invested and NOPAT, cash flow available to investors can be calculated using two methodologies: one focuses on how the cash flow is generated, and the other focuses on the recipients of free cash flow. Although the two methods seem redundant,

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EXHIBIT 11.3 **An Example of Free Cash Flow**

\$ million			
Accountant's cash flow statement		Free cash flow	
	Current year		Current year
Net income	198	NOPAT	210
Depreciation	20	Depreciation	20
Decrease (increase) in inventory	(25)	Gross cash flow	230
Increase (decrease) in accounts payable	25		
Cash flow from operations	218	Decrease (increase) in operating cash	(10)
		Decrease (increase) in inventory	(25)
Capital expenditures	(70)	Increase (decrease) in accounts payable	25
Decrease (increase) in equity investments	(10)	Capital expenditures	(70)
Cash flow from investing	(80)	Free cash flow	150
Increase (decrease) in interest-bearing debt	(25)	Nonoperating income	4
Dividends	(103)	Nonoperating taxes	4
Cash flow from financing	(128)	Decrease (increase) in equity investments	(10)
Starting cash	5	Cash flow available to investors	148
Cash flow from operations	218		
Cash flow from investing	(80)		
Cash flow from financing	(128)		
Ending cash	15		
		Reconciliation of cash flow available to investors	
		Interest expense	20
		Increase (decrease) in interest-bearing debt	25
		Dividends	103
		Cash flow available to investors	148

Subtract investments in operating items from gross cash flow

Evaluate cash flow from nonoperating assets separately from free cash flow

Treat interest as a financial payout to investors, not an expense

checking that both give the same result can help avoid line item omissions and classification pitfalls.

REORGANIZING THE ACCOUNTING STATEMENTS: IN PRACTICE

Reorganizing a company's financial statements can be difficult, even for the savviest analyst. Which assets are operating assets? Which are nonoperating? Which liabilities should be treated as debt? Which count as equity?

In the following pages, we examine reorganization in practice using Costco Wholesale. (A complete valuation of Costco with commentary is presented in Appendix H.) Costco, the fourth-largest retailer in the world, is well known for selling everyday items in bulk. It has stores in Australia, Canada, Iceland, Japan, Mexico, South Korea, the United Kingdom, and the United States. The company entered China in 2019. We set the stage for analyzing Costco's financial performance by first reorganizing its financial statements into operating, nonoperating, and financial items.

EXHIBIT 11.4 **Costco: Balance Sheet**

\$ million

Assets	2015	2016	2017	2018	2019
Cash and cash equivalents ¹	6,419	4,729	5,779	7,259	9,444
Receivables, net	1,224	1,252	1,432	1,669	1,535
Merchandise inventories	8,908	8,969	9,834	11,040	11,395
Deferred income taxes ²	521	—	—	—	—
Other current assets	227	268	272	321	1,111
Total current assets	17,299	15,218	17,317	20,289	23,485
Property, plant, and equipment	15,401	17,043	18,161	19,681	20,890
Deferred income taxes ²	109	202	254	316	398
Other assets	631	700	615	544	627
Total assets	33,440	33,163	36,347	40,830	45,400
Liabilities and shareholders' equity					
Accounts payable	9,011	7,612	9,608	11,237	11,679
Accrued salaries and benefits	2,468	2,629	2,703	2,994	3,176
Accrued member awards	813	869	961	1,057	1,180
Deferred membership fees	1,269	1,362	1,498	1,624	1,711
Current portion of long-term debt	1,283	1,100	86	90	1,699
Current portion of capital leases ³	10	10	7	7	26
Other current liabilities	1,686	1,993	2,632	2,917	3,766
Total current liabilities	16,540	15,575	17,495	19,926	23,237
Long-term debt	4,864	4,061	6,573	6,487	5,124
Capital leases ³	286	364	373	390	395
Deferred income taxes ²	462	297	312	317	543
Other liabilities	445	534	515	607	517
Total liabilities	22,597	20,831	25,268	27,727	29,816
Costco shareholders' equity	10,617	12,079	10,778	12,799	15,243
Noncontrolling interests	226	253	301	304	341
Total shareholders' equity	10,843	12,332	11,079	13,103	15,584
Liabilities and shareholders' equity	33,440	33,163	36,347	40,830	45,400

Note: Costco's fiscal year ends on the Sunday nearest August 31. For example, FY 2019 ended on September 1, 2019.

¹ Includes short-term investments.

² Deferred taxes are aggregated in other current assets, other assets, and other liabilities in original filings.

³ Capital leases are aggregated in other current liabilities and other liabilities in original filings.

Invested Capital: In Practice

To compute invested capital, we reorganize the company's balance sheet. Exhibit 11.4 presents historical balance sheets for Costco, whose fiscal year ends on the Sunday nearest August 31. The version presented is slightly more detailed than the balance sheets reported in Costco's annual reports, because we have searched the notes in each annual report for information about accounts that mix operating and nonoperating items. For instance, the notes in

Costco's 2019 annual report reveal that the company aggregates capital leases in other liabilities. Since capital leases are a form of debt and must be treated as such, the balance sheet in its original form would be unusable for valuation purposes.

Invested capital combines operating working capital (current operating assets minus current operating liabilities), fixed assets (net property, plant, and equipment), net other long-term operating assets (net of long-term operating liabilities), and when appropriate, intangible assets (goodwill, acquired intangibles, and capitalized software). Exhibit 11.5 demonstrates this line-by-line aggregation for Costco. In the following subsections, we examine each element in detail.

Operating Working Capital Operating working capital represents operating current assets minus operating current liabilities. Operating current assets comprise all current assets necessary for the operation of the business, including working cash balances, trade accounts receivable, inventory, and prepaid expenses. Specifically *excluded* are excess cash and marketable securities—that is, cash greater than the operating needs of the business.¹ Excess cash generally represents temporary imbalances in the company's cash position. We discuss this later in this section.²

Operating current liabilities include those liabilities that are related to the ongoing operations of the firm. The most common operating liabilities are those related to suppliers (accounts payable), employees (accrued salaries), customers (as either prepayments or, in the case of Costco, deferred membership fees), and the government (income taxes payable).³ If a liability is deemed operating rather than financial, it should be netted from operating assets to determine invested capital and consequently incorporated into free cash flow. Interest-bearing liabilities are nonoperating and should *not* be netted from operating assets, but rather valued separately (the related interest expense is classified as a nonoperating expense).

Some argue that operating liabilities, such as accounts payable, are a form of financing and should be treated no differently than debt. However, this

¹ Analyze excess cash separately from operating working capital for two reasons. First, excess cash is more accurately valued using a market value rather than as part of free cash flow. Second, excess cash will have a much lower risk–return profile than operating capital. Commingling assets with different risk profiles can distort your perception of performance.

² In a company's financial statements, accountants often distinguish between cash and marketable securities, but not between working cash and excess cash. We provide guidance on distinguishing working cash from excess cash later in this chapter.

³ When analyzing Costco, we treat accrued member rewards as an operating-related current liability and thus part of working capital. While we believe accrued member rewards are no different from other customer prepayments, the member is not paying cash specifically for the reward. One alternative is to use cash accounting for accrued member rewards, treating the liability as an equity equivalent. To convert to cash, add the increase in accrued member rewards to EBITA. Since taxes will not change, compute taxes using original EBITA.

EXHIBIT 11.5 **Costco: Invested Capital and Total Funds Invested**

\$ million

	2015	2016	2017	2018	2019
Operating cash ¹	2,324	2,374	2,581	2,832	3,054
Receivables, net	1,224	1,252	1,432	1,669	1,535
Merchandise Inventories	8,908	8,969	9,834	11,040	11,395
Other current assets	227	268	272	321	1,111
Operating current assets	12,683	12,863	14,119	15,862	17,095
Accounts payable	(9,011)	(7,612)	(9,608)	(11,237)	(11,679)
Accrued salaries and benefits	(2,468)	(2,629)	(2,703)	(2,994)	(3,176)
Accrued member awards	(813)	(869)	(961)	(1,057)	(1,180)
Deferred membership fees	(1,269)	(1,362)	(1,498)	(1,624)	(1,711)
Other current liabilities	(1,686)	(1,993)	(2,632)	(2,917)	(3,766)
Operating current liabilities	(15,247)	(14,465)	(17,402)	(19,829)	(21,512)
Operating working capital	(2,564)	(1,602)	(3,284)	(3,967)	(4,417)
Property, plant, and equipment	15,401	17,043	18,161	19,681	20,890
Capitalized operating leases ²	2,230	2,320	2,528	2,500	2,414
Other assets ³	631	700	615	544	627
Other liabilities ³	(445)	(534)	(515)	(607)	(517)
Invested capital	15,253	17,928	17,506	18,151	18,997
Excess cash ¹	4,095	2,355	3,199	4,427	6,390
Foreign tax credit carryforward ⁴	—	—	—	—	65
Total funds invested	19,348	20,282	20,704	22,578	25,452
Reconciliation of total funds invested					
Long-term debt and capital leases ⁵	6,443	5,535	7,039	6,974	7,244
Capitalized operating leases ²	2,230	2,320	2,528	2,500	2,414
Debt and debt equivalents	8,673	7,855	9,567	9,474	9,658
Deferred income taxes, operating ⁴	(61)	158	76	(39)	120
Deferred income taxes, nonoperating ⁴	(107)	(63)	(18)	40	90
Noncontrolling interests	226	253	301	304	341
Costco shareholders' equity	10,617	12,079	10,778	12,799	15,243
Equity and equity equivalents	10,675	12,427	11,137	13,104	15,794
Total funds invested	19,348	20,282	20,704	22,578	25,452

¹ Operating cash estimated at 2% of revenues. Remaining cash is treated as excess cash.² Capitalized operating leases are estimated for 2019 in Exhibit 22.10.³ Other assets and liabilities are classified as operating because no description is provided by the company.⁴ Foreign tax credit carryforward and other deferred taxes are reported in Exhibit 11.7.⁵ Includes current portion.

would lead to a definition of NOPAT that is inconsistent with invested capital. NOPAT is the income available to both debt and equity holders, so when you are determining ROIC, you should divide NOPAT by debt plus equity. Although a supplier may charge customers implicit interest for the right to pay in 30 days, the charge is an indistinguishable part of the price, and hence an indistinguishable and inseparable part of the cost of goods sold. Since cost of

goods sold is subtracted from revenue to determine NOPAT, operating liabilities must be subtracted from operating assets to determine invested capital. A theoretical but cumbersome alternative would be to treat accounts payable as debt and adjust NOPAT for the implicit interest cost.

Property, Plant, Equipment, and Other Capitalized Investments Include the book value of property, plant, and equipment net of accumulated depreciation in operating assets. Book value measures the company's ability to create value on past investments. Use market value or replacement cost only when evaluating the sale or replacement of a specific asset.

Some companies, including IBM and UPS, have significant investments in software they have developed for internal use. Under certain restrictions, these investments can be capitalized on the balance sheet rather than immediately expensed. Although it is labeled as an intangible asset, treat capitalized software no differently than property and equipment; treat amortization as if it were depreciation; and treat investments in capitalized software as if they were capital expenditures. (The cash flow statement in the IBM annual report separates investment in software from investment in PP&E. In contrast, UPS combines the two accounts within capital expenditures. In this case, attributing reported capitalized expenditures entirely to PP&E would overstate the actual investment.) Only internally generated intangible assets, and not acquired intangibles, should be treated in this manner. Acquired intangibles require special care and are discussed in a later subsection in this chapter.

Other Operating Assets, Net of Liabilities If other long-term assets and liabilities are small—and not detailed by the company—we typically assume they are operating. To determine net other long-term operating assets, subtract other long-term liabilities from other long-term assets. This figure should be included as part of invested capital. If, however, other long-term assets and liabilities are relatively large, you will need to disaggregate each account into its operating and nonoperating components before you can calculate other long-term operating assets, net of other liabilities.

For instance, a relatively large other long-term assets account might include nonoperating items such as deferred-tax assets, prepaid pension assets, nonconsolidated subsidiaries, or other equity investments. Nonoperating items should not be included in invested capital. Classifying assets as operating or nonoperating requires judgment, especially for obscure accounts. For instance, we treat restricted cash as operating when cash must be set aside to secure third-party guarantees, as is the case with distressed airlines that accept credit card payments with payment insurance. As a helpful guidepost, operating assets typically scale with revenues.

Long-term liabilities might similarly include operating and nonoperating items. Operating liabilities are liabilities that result directly from an ongoing

operating activity. For instance, one manufacturer records long-term customer advances within other liabilities. In general, however, most long-term liabilities are not operating liabilities, but rather what we deem debt and equity equivalents. These include unfunded pension liabilities, unfunded postretirement medical costs, restructuring reserves, and deferred taxes.

Where can you find a breakdown of other assets and other liabilities in the annual report? In some cases, companies provide a comprehensive table in the footnotes. Most of the time, however, you must work through the footnotes, note by note, searching for items aggregated within other assets and liabilities.

Goodwill and Acquired Intangibles In Chapter 12, return on invested capital is analyzed both with and without goodwill and acquired intangibles. ROIC with goodwill and acquired intangibles measures a company's ability to create value after paying acquisition premiums. ROIC without goodwill and acquired intangibles measures the competitiveness of the underlying business. For example, our colleagues studied the return on capital for large consumer packaged-goods companies from 1963 through 2009. What they found was intriguing. From the 1960s through the mid-1980s, the median ROIC without goodwill of these companies was consistently in the mid-teens. ROIC with goodwill was only slightly lower. Then, beginning in the mid-1980s, the companies were able to use the power of their brands to increase their ROIC without goodwill to a median of almost 35 percent. At the same time, they also stepped up their acquisition activity. Their median ROIC including goodwill remained in the mid to high teens. By 2009, the gap between the ROIC with goodwill and ROIC without goodwill was 17 percentage points. When you are analyzing the performance of a company, it's critical to understand ROIC with and without goodwill.

To evaluate the effect of goodwill and acquired intangibles properly, you should make two adjustments. First, subtract deferred-tax liabilities related to the amortization of acquired intangibles.⁴ Why? When amortization is not tax deductible, accountants create a deferred-tax liability at the time of the acquisition that is drawn down over the amortization period (since reported taxes will be lower than actual taxes). To counterbalance the liability, acquired intangibles are artificially increased by a corresponding amount, even though no cash is laid out. Subtracting deferred taxes related to acquired intangibles eliminates this distortion. For companies with significant acquired intangibles—for example, Coca-Cola—the adjustment can be substantial.

Second, add back cumulative amortization and impairment. Unlike other fixed assets, goodwill and acquired intangibles do not wear out, nor are they replaceable. Therefore, you need to adjust reported goodwill and acquired

⁴Since goodwill is tested regularly for impairment and cannot be amortized, this issue relates only to acquired intangibles.

intangibles upward to recapture historical impairments of goodwill and amortization of intangibles. (To maintain consistency, do not deduct impairments of goodwill or amortization of acquired intangibles from revenues to determine NOPAT. This is why NOPAT starts with EBITA.)

Consider FedEx, which wrote down approximately \$900 million in goodwill and acquired intangibles when it converted the acquired brand name Kinko's to FedEx Office. Failing to add back this impairment would have caused a large artificial jump in return on invested capital following the write-down. The money spent on an acquisition is real and needs to be accounted for, even when the investment loses value.

Computing Total Funds Invested

Invested capital represents the capital necessary to operate a company's core business. In addition to invested capital, companies can also own nonoperating assets. The combination of invested capital and nonoperating assets leads to total funds invested. Nonoperating assets include excess cash and marketable securities, receivables from financial subsidiaries (for example, credit card receivables), nonconsolidated subsidiaries, overfunded pension assets, and tax loss carry-forwards. Costco has two nonoperating assets: excess cash and foreign tax credit carryforwards.

There are two reasons to diligently separate operating and nonoperating assets. First, nonoperating assets can distort performance measures for both economic and accounting reasons. For example, many nonoperating assets generate income, but companies do not report the income unless certain ownership thresholds are met. Including an asset without its corresponding income distorts performance measurements. Second, there are better methods than discounted cash flow to value nonoperating assets. You would never discount interest income to value excess cash. For this asset, the book value suffices.

Now let's examine the most common nonoperating assets.

Excess Cash and Marketable Securities Do not include excess cash in invested capital. By its definition, excess cash is unnecessary for core operations. Rather than mix excess cash with core operations, analyze and value excess cash separately. Given its liquidity and low risk, excess cash will earn very small returns. Failing to separate excess cash from core operations will incorrectly depress the company's apparent ROIC.

Companies do not disclose how much cash they deem necessary for operations. Nor does the accounting definition of cash versus marketable securities distinguish working cash from excess cash. Based on past analysis, companies with the smallest cash balances held cash just below 2 percent of sales. If this is a good proxy for working cash, any cash above 2 percent

should be considered excess.⁵ In 2019, Costco held just under \$9.5 billion in cash and marketable securities on \$152.7 billion in revenue. At 2 percent of revenue, operating cash equals \$3.1 billion. The remaining cash of \$6.4 billion is treated as excess. Exhibit 11.5 separates operating cash from excess cash. Excess cash is not included in invested capital, but rather is treated as a non-operating asset.

Nonconsolidated Subsidiaries and Equity Investments Nonconsolidated subsidiaries, also referred to as investments in associates, investments in affiliated companies, and equity investments, should be measured and valued separately from invested capital. When a company owns a minority stake in another company, it will record the investment as a single line item on the balance sheet and will not record the individual assets owned by the subsidiary. On the income statement, only the net income from the subsidiary will be recorded on the parent's income statement, not the subsidiary's revenues or costs. Since only net income—not revenue—is recorded, including nonconsolidated subsidiaries as part of operations will distort margins and capital turnover. Therefore, we recommend separating nonconsolidated subsidiaries from invested capital and analyzing and valuing nonconsolidated subsidiaries separately from core operations.

Financial Subsidiaries Some companies, including General Motors and Siemens, have financing subsidiaries that finance customer purchases. Because these subsidiaries charge interest on financing for purchases, they resemble banks. Since bank economics are quite different from those of manufacturing and service companies, you should separate line items related to the financial subsidiary from the line items for the manufacturing business. Then evaluate the return on capital for each type of business separately. Otherwise, significant distortions of performance will make a meaningful comparison with competitors impossible. For more on how to analyze and assess financial subsidiaries, see Chapter 19.

Overfunded Pension Assets If a company runs a defined-benefit pension plan for its employees, it must fund the plan each year. And if a company funds its plan faster than its pension expenses dictate or assets grow faster than expected, under U.S. Generally Accepted Accounting Principles (GAAP) and International Accounting/Financial Reporting Standards (IAS/IFRS) the

⁵ This aggregate figure, however, is not a rule. Required cash holdings vary by industry. For instance, one study found that companies in industries with higher cash flow volatility hold higher cash balances. To assess the minimum cash needed to support operations, look for a minimum clustering of cash to revenue across the industry. To better understand the reason behind significant cash holdings in a historical context, see J. Graham and M. Leary, "The Evolution of Corporate Cash," SSRN working paper (May 25, 2018).

company can recognize a portion of the excess assets on the balance sheet. Pension assets are considered a nonoperating asset and not part of invested capital. Their value is important to the equity holder, so they will be valued later, but separately from core operations. Chapter 23 examines pension assets in detail.

Tax Loss Carryforwards Unless they are small and grow consistently with revenue, do not include tax loss carryforwards—also known as net operating losses (NOLs)—as part of invested capital. Depending on the type of deferred-tax asset, it will be valued either separately or as part of operating cash taxes. Given the complexity of reorganizing deferred taxes, we discuss them in more detail later in this chapter, in the subsection titled “Equity Equivalents Such as Deferred Taxes.”

Other Nonoperating Assets Other nonoperating assets, such as derivatives, excess real estate, and discontinued operations, also should be excluded from invested capital. For Costco, derivatives were disclosed in the footnotes but were immaterial, so no adjustments were made to the balance sheet accounts.

Reconciling Total Funds Invested

Total funds invested can be calculated as invested capital plus nonoperating assets, as in the previous section, or as the sum of debt, equity, and their equivalents. The totals produced by the two approaches should reconcile. A summary of sources of financing appears in Exhibit 11.6. We next examine each of these sources of capital contributing to total funds invested.

EXHIBIT 11.6 **Sources of Financing**

Source of capital	Description
Debt	Interest-bearing debt from banks and public capital markets
Debt equivalents	Off-balance-sheet debt and one-time debts owed to others that are not part of ongoing operations (e.g., severance payments as part of a restructuring, an unfunded pension liability, or expected environmental remediation following a plant closure)
Equity	Common stock, additional paid-in capital, retained earnings, and accumulated other comprehensive income
Equity equivalents	Balance sheet accounts that arise because of noncash adjustments to retained earnings; similar to debt equivalents but not deducted from enterprise value to determine equity value (e.g., most deferred-tax accounts and income-smoothing provisions)
Hybrid securities	Claims that have equity characteristics but are not yet part of owners' equity (e.g., convertible debt and employee options)
Noncontrolling interest by other companies	External shareholders' minority position in any of the company's consolidated subsidiaries

Debt Debt includes all short-term or long-term interest-bearing liabilities. Short-term debt includes commercial paper, notes payable, and the current portion of long-term debt. Long-term debt includes fixed debt, floating debt, and convertible debt with maturities of more than a year.

Debt Equivalents Such as Retirement Liabilities and Restructuring Reserves

If a company's defined-benefit plan is underfunded, it must recognize the underfunding as a liability. The amount of underfunding is not an operating liability. Rather, treat unfunded pension liabilities and unfunded postretirement medical liabilities as a debt equivalent (and treat the net interest expense associated with these liabilities as nonoperating). It is as if the company must borrow money to fund the plan. As an example, UPS announced in 2012 that it would withdraw from a multiemployer pension fund. To be released from its obligations to the fund, UPS promised to pay \$43 million per year for 50 years. This fixed repayment promise, an obligation with seniority to equity claims, is no different from traditional debt.

We discuss other debt equivalents, such as reserves for plant decommissioning and restructuring reserves, in Chapter 21.

Equity Equity includes original investor funds, such as common stock and additional paid-in capital, as well as investor funds reinvested into the company, such as retained earnings and accumulated other comprehensive income (OCI). In the United States, accumulated OCI consists primarily of currency adjustments, aggregate unrealized gains and losses from liquid assets whose value has changed but that have not yet been sold, and pension plan fluctuations within a certain band. IFRS also includes accumulated OCI within shareholders' equity but reports each reserve separately. Any stock repurchased and held in the treasury should be deducted from total equity. In Exhibit 11.5, we consolidate these accounts into a single account titled shareholders' equity.

Equity Equivalents Such as Deferred Taxes Equity equivalents are balance sheet accounts that arise because of noncash adjustments to retained earnings. Equity equivalents are like debt equivalents; they differ only in that they are not deducted from enterprise value to determine equity value.

The most common equity equivalent, deferred taxes, arises from differences in how businesses and the government account for taxes. For instance, the government typically uses accelerated depreciation to determine a company's taxes, whereas the accounting statements are prepared using straight-line depreciation. This leads to cash taxes that are lower than reported taxes during the early years of an asset's life. For growing companies, this difference will cause reported taxes consistently to overstate the company's actual tax payments. To avoid this bias, use cash-based (versus accrual) taxes to determine NOPAT. Since reported taxes will now match cash taxes on the income statement, the

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EXHIBIT 11.7 **Costco: Reorganized Deferred Taxes**

\$ million

	As reported				Reorganized		
	2017	2018	2019		2017	2018	2019
Deferred-tax assets				Operating deferred-tax assets, net of liabilities			
Equity compensation	109	72	74	Equity compensation	109	72	74
Deferred income/membership fees	167	136	180	Deferred income/membership fees	167	136	180
Foreign tax credit carryforward	—	—	65	Accrued liabilities and reserves	647	484	566
Accrued liabilities and reserves	647	484	566	Property and equipment	(747)	(478)	(677)
Other	18	—	—	Merchandise inventories	(252)	(175)	(187)
Total deferred-tax assets	941	692	885	Valuation allowance	—	—	(76)
				Operating deferred-tax assets, net of liabilities	(76)	39	(120)
Valuation allowance	—	—	(76)				
Total net deferred-tax assets	941	692	809	Nonoperating deferred-tax assets, net of liabilities			
Deferred-tax liabilities				Other assets	18	—	—
Property and equipment	(747)	(478)	(677)	Foreign branch deferreds	—	—	(69)
Merchandise inventories	(252)	(175)	(187)	Other liabilities	—	(40)	(21)
Foreign branch deferreds	—	—	(69)	Nonoperating deferred-tax assets, net of liabilities	18	(40)	(90)
Other	—	(40)	(21)				
Total deferred-tax liabilities	(999)	(693)	(954)	Tax loss carryforwards			
				Foreign tax credit carryforward	—	—	65
Deferred-tax assets, net of liabilities	(58)	(1)	(145)	Deferred-tax assets, net of liabilities	(58)	(1)	(145)

deferred-tax account—in this case related to accelerated depreciation—is no longer necessary. This is why the deferred-tax account is referred to as an equity equivalent. It represents the adjustment to retained earnings that would be made if the company reported cash taxes to investors instead of accrual taxes.

Not every deferred-tax account is operating. Although both operating and nonoperating deferred-tax accounts are equity equivalents, incorporate only deferred-tax accounts associated with ongoing operations into operating cash taxes.⁶ In contrast, value nonoperating deferred taxes as part of the corresponding account.⁷ For instance, when valuing an underfunded pension, do not use the book value of deferred taxes to value potential tax savings. Instead, reduce the underfunding by the projected taxes likely to be saved when the plan is funded.

Exhibit 11.7 converts deferred-tax assets and liabilities for Costco into operating, nonoperating, and tax loss carryforwards, using the tax footnote in the company's annual report. Although individual operating-related accounts, such as accrued liabilities and reserves, are large, the net amount is close to zero. For this reason, operating cash taxes for Costco will not differ significantly from accrual-based taxes.

⁶ Separating deferred taxes into operating and nonoperating items can be challenging and often requires advanced knowledge of accounting conventions. For an in-depth discussion of deferred taxes, see Chapter 20.

⁷ As discussed earlier, deferred-tax assets related to past losses should be classified as a nonoperating asset and valued separately. Deferred-tax liabilities related to amortization of acquired intangibles should be netted against acquired intangibles. These accounts are not equity equivalents.

Hybrid Securities and Noncontrolling Interests Some sources of financing resist easy classification as debt or equity. These include hybrid securities and noncontrolling interests. Unlike debt, these accounts do not have fixed interest payments. Unlike equity, they are not the residual claim on cash flows. Therefore, these accounts should be valued separately and deducted from enterprise value to determine equity value.

- *Hybrid securities.* The three most common hybrid securities are convertible debt, preferred stock, and employee options. Since hybrid securities contain embedded options, they cannot be treated as common stock. Instead, use the market price or, if necessary, option-pricing models to value these claims separately. Failing to do so can undervalue the hybrid security and overstate the value of common stock. This is especially important for venture-capital-backed preferred stock and long-dated employee options.
- *Noncontrolling interests.* A noncontrolling interest occurs when a third party owns a minority holding in one of the company's consolidated subsidiaries. If a noncontrolling interest exists, treat the balance sheet amount as a source of financing. Treat the earnings attributable to any noncontrolling interest as a financing cash flow similar to dividends. If data are available, value the subsidiary separately, and deduct the noncontrolling interest from the company's enterprise value to determine equity value. If data for the subsidiary are available, discount earnings related to the noncontrolling interest at an appropriate cost of equity. Chapter 16 presents various valuation methodologies for noncontrolling interests.

Correctly classifying balance sheet items can be a daunting task. But fret not: perfect classification is not required. You need only to assure that each account is included as part of free cash flow or valued separately.

Calculating NOPAT

To determine NOPAT for Costco, we turn to the income statement (see Exhibit 11.8) and convert it into NOPAT, as shown in Exhibit 11.9.

Net Operating Profit (EBITA) NOPAT starts with earnings before interest, taxes, and amortization (EBITA) of acquired intangibles, which equals revenue minus operating expenses, such as cost of goods sold, selling costs, general and administrative costs, and depreciation.

Why use EBITA and not EBITDA? When a company purchases a physical asset such as equipment, it capitalizes the asset on the balance sheet and depreciates the asset over its lifetime. Since the asset wears out over time,

EXHIBIT 11.8 **Costco: Income Statement**

\$ million

	2015	2016	2017	2018	2019
Merchandise sales	113,666	116,073	126,172	138,434	149,351
Membership fees	2,533	2,646	2,853	3,142	3,352
Revenues	116,199	118,719	129,025	141,576	152,703
Merchandise costs	(101,065)	(102,901)	(111,882)	(123,152)	(132,886)
Selling, general, and administrative	(10,318)	(10,813)	(11,580)	(12,439)	(13,502)
Depreciation ¹	(1,127)	(1,255)	(1,370)	(1,437)	(1,492)
Preopening expenses	(65)	(78)	(82)	(68)	(86)
Operating income	3,624	3,672	4,111	4,480	4,737
Interest expense	(124)	(133)	(134)	(159)	(150)
Interest income	50	41	50	75	126
Other income	54	39	12	46	52
Earnings before taxes	3,604	3,619	4,039	4,442	4,765
Provision for income taxes	(1,195)	(1,243)	(1,325)	(1,263)	(1,061)
Net income, consolidated	2,409	2,376	2,714	3,179	3,704
Net income, noncontrolling interests	(32)	(26)	(35)	(45)	(45)
Net income, Costco	2,377	2,350	2,679	3,134	3,659

¹ Aggregated in selling, general, and administrative expenses in original filings.

any measure of profit (and return) must recognize this loss in value. While depreciation does not match the periodic loss in value perfectly, it is a suitable proxy.

Why use EBITA and not EBIT? After all, the same argument could be made for the amortization of acquired intangibles: they, too, have fixed lives and lose value over time. But the accounting for intangibles differs from the accounting for physical assets. Unlike capital expenditures, internally created intangible assets such as new customer lists and product brands are *expensed* and not capitalized. Thus, when the acquired intangible loses value and is replaced through additional investment internally, the reinvestment is *already* expensed, and the company is penalized twice in the same time period: once through amortization and a second time through reinvestment. Although not perfect, using EBITA is consistent with existing accounting rules.

Choosing which line items to include as operating expenses requires judgment. As a guiding principle, include ongoing expenses related to the company's core operations. One company we recently analyzed included rationalizations as part of operating expenses. Since rationalizations had been a consistent part of the company's expense structure and are likely to continue as the industry continues to mature, we kept them as operating expenses. Had they been a one-time expense, we would not have included them in EBITA.

EXHIBIT 11.9 **Costco: NOPAT and Its Reconciliation to Net Income**

\$ million

	2015	2016	2017	2018	2019
Revenue	116,199	118,719	129,025	141,576	152,703
Merchandise costs	(101,065)	(102,901)	(111,882)	(123,152)	(132,886)
Selling, general, and administrative	(10,318)	(10,813)	(11,580)	(12,439)	(13,502)
Depreciation	(1,127)	(1,255)	(1,370)	(1,437)	(1,492)
Preopening expenses	(65)	(78)	(82)	(68)	(86)
EBITA, unadjusted ¹	3,624	3,672	4,111	4,480	4,737
Operating lease interest ²	73	75	57	74	91
EBITA, adjusted	3,697	3,747	4,168	4,554	4,828
Operating cash taxes ³	(1,184)	(1,149)	(1,493)	(1,455)	(1,009)
NOPAT	2,513	2,598	2,675	3,098	3,818
Reconciliation to net income					
Net income, consolidated	2,409	2,376	2,714	3,179	3,704
Operating taxes deferred ³	7	219	(82)	(115)	159
Adjusted net income	2,416	2,595	2,632	3,064	3,863
Interest expense	124	133	134	159	150
Operating lease interest ²	73	75	57	74	91
Interest income	(50)	(41)	(50)	(75)	(126)
Other income ⁴	(54)	(39)	(12)	(46)	(52)
Taxes related to nonoperating accounts ⁵	(35)	(48)	(49)	(32)	(15)
Other nonoperating taxes ³	39	(77)	(37)	(45)	(92)
NOPAT	2,513	2,598	2,675	3,098	3,818

¹ Earnings before interest, taxes, and amortization.² Operating lease interest is estimated in Exhibit 11.15.³ Operating cash taxes and other nonoperating taxes are detailed in Exhibit 11.11.⁴ Other income consists primarily of foreign-currency transaction gains and treated as nonoperating for simplicity of exposition.⁵ Estimated by multiplying the statutory tax rate by the sum of interest and operating lease interest expense, less the sum of interest and other income. The statutory tax rate is reported in Exhibit 11.10.

Adjustments to EBITA In many companies, nonoperating items are embedded within operating expenses. To ensure that your EBITA calculation flows solely from operations, dig through the notes to weed out nonoperating items from operating expenses. The most common nonoperating items are related to pensions, embedded interest expenses from operating leases, and one-time restructuring charges hidden in the cost of sales.

In Exhibit 11.9, we adjust operating profit for operating lease interest. Since Costco does not offer defined-benefit retirement plans, no adjustment was made for the nonoperating portion of pension expense. The comprehensive processes for operating leases and pensions are addressed at the end of this chapter and in Part Three of this book, which covers advanced valuation issues.

No other adjustments were required, as Costco did not embed material one-time items in operating expenses. Although not common, it can happen.

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EXHIBIT 11.10 **Costco: Tax Reconciliation Table**

\$ million

	2015	2016	2017	2018	2019
Federal taxes at statutory rate	1,262	1,267	1,414	1,136	1,001
State taxes, net	85	91	116	154	171
Foreign taxes, net	(125)	(21)	(64)	32	(1)
Employee stock ownership plan (ESOP)	(66)	(17)	(104)	(14)	(18)
2017 tax act	—	—	—	19	(123)
Other	39	(77)	(37)	(64)	31
U.S. and foreign tax expense (benefit)	1,195	1,243	1,325	1,263	1,061
Tax rates¹					
Federal income tax rate, %	35.0	35.0	35.0	25.6	21.0
State income tax rate, %	2.4	2.5	2.9	3.5	3.6
Statutory tax rate, %	37.4	37.5	37.9	29.0	24.6

¹ To determine each tax rate, divide each tax amount by earnings before taxes. Earnings before taxes are reported in Exhibit 11.8.

Source: Reported in Costco's annual report, note 8: Income Taxes.

UPS's decision to withdraw from a multiemployer pension plan in 2012 caused its compensation and benefits expense to spike that year. Since the withdrawal was a one-time event, it is better evaluated separately as a nonoperating expense and not embedded in operating income. Choosing whether an expense is one-time or ongoing requires judgment. Separating one-time items from ongoing expenses, however, highlights trends and opens the valuation discussion to future risks.

Operating Cash Taxes Since many nonoperating items affect income taxes, they also must be adjusted to an all-equity operating level. The process for adjusting taxes is the most complicated part of reorganizing the financial statements. Chapter 20 goes into more detail about the specifics of the process, the reasoning behind it, and alternative ways to implement it. For now, we summarize the process.

To determine operating taxes, you will need the tax reconciliation table from the company's notes. Some companies report the tax reconciliation table in percent; others report the table in currency. In Chapter 20, we present how to estimate operating taxes using both reporting styles. Exhibit 11.10 presents the tax reconciliation table for Costco.

To estimate operating cash taxes, proceed in three steps:

1. Using the tax reconciliation table, determine the statutory tax rate. The statutory tax rate equals the government tax rate paid on income. Multiply the statutory tax rate by adjusted EBITA to determine statutory taxes on adjusted EBITA.
2. Increase (or decrease) statutory taxes on EBITA by other operating taxes (or credits). To estimate other operating taxes, search the tax reconciliation table for ongoing, operating-related taxes other than statutory taxes. The most

EXHIBIT 11.11 **Costco: Taxes**

\$ million

	2015	2016	2017	2018	2019
EBITA	3,697	3,747	4,168	4,554	4,828
× Statutory tax rate ¹	37.4%	37.5%	37.9%	29.0%	24.6%
Statutory taxes on EBITA	1,382	1,406	1,579	1,322	1,187
Foreign taxes, net ²	(125)	(21)	(64)	32	(1)
Employee stock ownership plan (ESOP) ²	(66)	(17)	(104)	(14)	(18)
Operating taxes	1,191	1,368	1,411	1,340	1,168
Operating taxes deferred ³	(7)	(219)	82	115	(159)
Operating cash taxes	1,184	1,149	1,493	1,455	1,009
Reported taxes					
Operating taxes	1,191	1,368	1,411	1,340	1,168
Taxes related to nonoperating accounts ⁴	(35)	(48)	(49)	(32)	(15)
Other nonoperating taxes ⁵	39	(77)	(37)	(45)	(92)
Income taxes, reported	1,195	1,243	1,325	1,263	1,061

¹ Estimated by dividing federal plus state income taxes by earnings before taxes.² Reported in the tax reconciliation table presented in Exhibit 11.10.³ Computed as the increase (decrease) in operating deferred tax assets, net of liabilities. Operating deferred taxes are reported in Exhibit 11.7.⁴ Estimated in Exhibit 11.9.⁵ Other nonoperating taxes include taxes related to the 2017 tax act and other taxes, reported in Exhibit 11.10.

common operating tax is the difference between domestic and foreign tax rates. Sum the other rates deemed operating, and if the table is presented in percent, multiply the resulting summation of by earnings before taxes (EBT). Multiplying the percentages by EBT (not EBITA) converts the percentages found in the tax reconciliation table into a dollar-based adjustment.⁸

3. Convert accrual-based taxes into operating cash taxes. For companies that systematically defer taxes, accrual-based taxes will not properly represent cash taxes actually paid. The simplest way to calculate operating cash taxes is to subtract the increase in operating deferred-tax liabilities (net of assets) from operating taxes. While the notes provide information on taxes that have been deferred, they do not separate operating from nonoperating deferred taxes, making the disclosure unusable. Not every company discloses enough information to separate operating deferred taxes, such as accelerated depreciation, from nonoperating deferred taxes, such as those related to prepaid pension assets. When this information is unavailable, we recommend using operating taxes without a cash adjustment.

To demonstrate the three-step process, we construct operating cash taxes for Costco in Exhibit 11.11. In 2019, the statutory tax rate for Costco was 24.6

⁸ When adjusting statutory taxes on EBITA for other operating items, we prefer to use dollar adjustments rather than percentage adjustments. This is because artificially low earnings before taxes can distort the percentages in a significant way. For instance, when UPS withdrew from the multistate pension plan in 2012, adjustment percentages related to foreign tax savings were uncharacteristically large because of the smaller-than-usual EBT.

percent. This value includes both federal taxes (21.0 percent) and state taxes (3.6 percent). To determine statutory taxes on EBITA, multiply the statutory tax rate (24.6 percent) by EBITA (\$4,828 million), which was estimated in Exhibit 11.9. In 2019, statutory taxes on EBITA were \$1,187 million.

Next, search the tax reconciliation table for other operating taxes. We classify foreign income taxed at rates different from the U.S. statutory rate (\$1 million) and tax savings from the employee stock ownership plan (\$18 million) as operating. In contrast, taxes related to the substantial change in U.S. corporate tax rates brought about by the 2017 Tax Cuts and Jobs Act are a one-time event. Therefore, treat them as nonoperating. To determine other operating taxes, sum across operating-related tax adjustments. In 2019, other operating taxes decreased Costco's taxes on EBITA by \$19 million. Summing statutory taxes on EBITA (\$1,187 million) and other operating taxes (−\$19 million) leads to \$1,168 million in operating taxes.

To convert operating taxes into operating cash taxes, add (subtract) the increase in *operating* deferred-tax assets (liabilities). As discussed in the section on invested capital, do not incorporate the change in nonoperating deferred taxes into cash taxes. Instead, value nonoperating deferred taxes as part of your valuation of the corresponding nonoperating account. For instance, future taxes on pension shortfalls should be computed using projected contributions, not on the historical deferred-tax account.

Exhibit 11.7 separates Costco's operating and nonoperating deferred taxes. Since operating deferred-tax assets net of liabilities decreased in 2019, Costco is paying less in cash taxes than reported using accrual accounting. In 2019, operating deferred-tax assets net of liabilities fell by \$159 million. Therefore, operating taxes of \$1,168 million is reduced by \$159 million to estimate operating cash taxes at \$1,009 million.⁹

Like other balance sheet accounts, operating deferred-tax accounts rise and fall for reasons other than deferrals, such as acquisitions, divestitures, and revaluations. However, only organic changes in deferred taxes should be included in operating cash taxes, not one-time changes resulting from revaluation or consolidation. For instance, most American companies revalued their 2018 deferred-tax accounts to reflect the 2017 Tax Cuts and Jobs Act. To estimate the organic change in deferred-tax assets and liabilities, estimate what the change would have been if tax rates had remained unchanged. In the case of Costco, the effect was immaterial.

For many companies, a clean measure of operating cash taxes may be impossible to calculate. When this is the case, use operating taxes without converting to cash.

⁹ In Appendix H, we forecast the operating cash tax rate as part of our valuation of Costco. Since the percentage of Costco's taxes that are deferred is volatile, we use a five-year average to estimate the percentage of operating taxes that are likely to be deferred.

Reconciliation of Reported Taxes To reconcile NOPAT to net income, it is helpful to first reconcile operating taxes to reported taxes. At the bottom of Exhibit 11.11, we present a reconciliation of reported taxes. The reconciliation includes the taxes related to nonoperating accounts and other nonoperating taxes. Although the two accounts sound similar, they are estimated differently.

The taxes related to nonoperating accounts, which equal $-\$15$ million in 2019, is calculated by multiplying the marginal tax rate by the sum of nonoperating accounts reported in the reconciliation of NOPAT to net income presented in Exhibit 11.9. For Costco, nonoperating accounts include interest expense, operating lease interest, interest income, and other income. To determine other nonoperating taxes, search the tax reconciliation table presented in Exhibit 11.10 for nonoperating items, such as one-time audits and write-offs. In the previous section, we classified taxes related to the 2017 Tax Cuts and Jobs Act ($-\$123$ million) and the “other” account ($\$31$ million) as nonoperating. Summing the two equals $-\$92$ million.

Note how the reconciliation ties to the reported income taxes on the income statement presented in Exhibit 11.8. Although reconciliation can be time-consuming, it assures that the modeling has been carried out correctly.

Reconciliation to Net Income

To ensure that the reorganization is accurate, we recommend reconciling net income to NOPAT (see the lower half of Exhibit 11.9). To reconcile NOPAT, start with net income available to both common shareholders and noncontrolling interests, and add back the increase (or subtract the decrease) in operating deferred-tax liabilities. We label this amount adjusted net income.

Next, add any nonoperating charges (or subtract any income) reported by the company, such as interest expense and other nonoperating expenses. After this, include any adjustments that have been made, like adjustments for operating lease interest and, if required, the nonoperating portion of the pension expense. Finally, subtract tax shields on the nonoperating expenses calculated previously and add any nonoperating taxes from the tax reconciliation table. Whether NOPAT is estimated using revenues less expenses or alternatively as net income plus nonoperating items and other adjustments, the result should be identical.

Free Cash Flow: In Practice

This subsection details how to build free cash flow from the reorganized financial statements. For estimating free cash flow, the income statement and balance sheet will not suffice; the statement of shareholders’ equity also is required. Exhibit 11.12 presents the statement of shareholders’ equity for Costco. This statement reconciles the income statement with the balance sheet and

EXHIBIT 11.12 **Costco: Statement of Shareholders' Equity**

\$ million

	2015	2016	2017	2018	2019
Equity, beginning of year	12,303	10,617	12,079	10,778	12,799
Net income	2,377	2,350	2,679	3,134	3,659
Foreign-currency translation adjustment	(1,045)	22	85	(185)	(237)
Comprehensive income	1,332	2,372	2,764	2,949	3,422
Stock-based compensation	394	459	518	547	598
Stock options exercised	69	—	—	—	—
Release of vested restricted stock units	(122)	(146)	(165)	(217)	(272)
Repurchases of common stock	(494)	(477)	(473)	(322)	(247)
Cash dividends declared	(2,865)	(746)	(3,945)	(936)	(1,057)
Equity, end of year	10,617	12,079	10,778	12,799	15,243

presents additional information required to estimate free cash flow and cash flow available to investors. Free cash flow is defined as:

$$\text{FCF} = \text{NOPAT} + \text{Noncash Operating Expenses} - \text{Investments in Invested Capital}$$

Exhibit 11.13 presents the free cash flow calculation for Costco and reconciles free cash flow to cash flow available to investors. To create free cash flow, start with NOPAT and add back noncash expenses, such as depreciation and depletion. From gross cash flow, subtract investments in working capital, capital expenditures, and investments in other long-term assets net of liabilities.

Gross Cash Flow Gross cash flow represents the cash operating profits that the company generates. It represents the cash available for investment and investor payout without the company having to sell nonoperating assets, such as excess cash, or to raise additional capital. Gross cash flow has two components:

1. *NOPAT*. As previously defined, net operating profit after taxes is the after-tax operating profit available to all investors.
2. *Noncash operating expenses*. Some expenses embedded in NOPAT are noncash and represent the economic decay of past investments. To convert NOPAT into cash flow, add back depreciation, depletion, and amortization of capitalized assets. Only add back amortization deducted from revenues to compute NOPAT, such as the amortization of capitalized software or purchased customer contracts. Do not add back the amortization from acquired intangibles and impairments to NOPAT;

EXHIBIT 11.13 **Costco: Free Cash Flow and Cash Flow to Investors**

\$ million

	2016	2017	2018	2019
NOPAT	2,598	2,675	3,098	3,818
Depreciation	1,255	1,370	1,437	1,492
Gross cash flow	3,853	4,045	4,535	5,310
Decrease (increase) in working capital	(962)	1,682	684	449
Less: Capital expenditures ¹	(2,649)	(2,502)	(2,969)	(2,998)
Decrease (increase) in capitalized operating leases	(91)	(208)	28	86
Decrease (increase) in other assets, net of liabilities	20	6	163	(173)
Free cash flow	171	3,083	2,441	2,675
Interest income	41	50	75	126
Other income	39	12	46	52
Taxes related to nonoperating accounts	48	49	32	15
Other nonoperating taxes	77	37	45	92
Decrease (increase) in excess cash	1,740	(844)	(1,229)	(1,962)
Decrease (increase) in tax credit carryforward	—	—	—	(65)
Unexplained foreign-currency translation ²	(226)	99	(173)	60
Cash flow to investors	1,890	2,486	1,238	993
Reconciliation of cash flow to investors				
Interest expense	133	134	159	150
Operating lease interest	75	57	74	91
Decrease (increase) in long-term debt and capital leases	908	(1,504)	65	(270)
Decrease (increase) in capitalized operating leases	(91)	(208)	28	86
Cash flow to debt and debt equivalents	1,025	(1,521)	326	57
Nonoperating deferred income taxes	(44)	(45)	(58)	(50)
Shares issued for stock-based compensation, net ³	(313)	(353)	(330)	(326)
Repurchases of common stock	477	473	322	247
Dividends	746	3,945	936	1,057
Payments to (investments in) noncontrolling interests ⁴	(1)	(13)	42	8
Cash flow to equity and equity equivalents	865	4,007	912	936
Cash flow to investors	1,890	2,486	1,238	993

¹ Capital expenditures are reported on the statement of cash flows.² Foreign-currency translation adjustment, less the portion allocated to the change of property, plant, and equipment; detailed in Exhibit 11.14.³ Includes stock-based compensation, stock options exercised, net of the release of vested restricted stock units.⁴ Equals net income to nonconsolidated interests minus (plus) the increase (decrease) in noncontrolling interests.

they were not subtracted from revenue in calculating NOPAT. Another major noncash expense is share-based employee compensation. Do not add back share-based compensation to NOPAT to determine gross cash flow. Since employees have a new claim on cash flows, this claim must be incorporated into the valuation, either as part of cash flow or as a separate calculation. (Share-based employee compensation is discussed in Chapter 16.)

Investments in Invested Capital To maintain and grow their operations, companies must reinvest a portion of their gross cash flow back into the business. To determine free cash flow, subtract gross investment from gross cash flow. We segment gross investment into five primary areas:

1. *Change in operating working capital.* Growing a business requires investment in operating cash, inventory, and other components of working capital. Operating working capital excludes nonoperating assets, such as excess cash, and financing items, such as short-term debt and dividends payable.
2. *Capital expenditures, net of disposals.* Capital expenditures represent investments in property, plant, and equipment (PP&E), less the book value of any PP&E sold. One way to estimate net capital expenditures is to add depreciation to the increase in net PP&E.¹⁰ Do not estimate capital expenditures by taking the change in gross PP&E. Since gross PP&E drops when companies retire assets, the change in gross PP&E will often understate the actual amount of capital expenditures.
3. *Change in capitalized operating leases.* To keep the definitions of NOPAT, invested capital, ROIC, and free cash flow consistent, include investments in capitalized operating leases in gross investment. Capitalized operating leases are discussed later in the chapter.
4. *Investment in goodwill and acquired intangibles.* For acquired intangible assets, where cumulative amortization has been added back, you can estimate investment by computing the change in net goodwill and acquired intangibles. For intangible assets that are being amortized, use the same method as for determining net capital expenditures (by adding amortization to the increase in net intangibles).
5. *Change in other long-term operating assets, net of long-term liabilities.* Subtract investments in other net operating assets. As with invested capital, do not confuse other long-term operating assets with other long-term nonoperating assets, such as equity investments and excess pension assets. Changes in nonoperating assets need to be evaluated—but should be analyzed separately.

For most assets and liabilities, the year-to-year change in a balance sheet account will suitably approximate net investment. This will not always be the case. Currency translations, acquisitions, write-offs, and accounting changes

¹⁰ If possible, use capital expenditures reported in the accounting statement of cash flows, but only after reconciling reported capital expenditures with the change of net PP&E plus depreciation. Capital expenditures can differ from net PP&E plus depreciation because of currency translations (discussed later in this section), acquisitions, and impairments. Acquisitions should be analyzed separately, and impairments should be treated as a nonoperating noncash expense in the income statement.

EXHIBIT 11.14 **Costco: Changes in Property, Plant, and Equipment**

\$ million

	2015	2016	2017	2018	2019
Property, plant, and equipment, beginning of year	14,830	15,401	17,043	18,161	19,681
Capital expenditures ¹	2,393	2,649	2,502	2,969	2,998
Depreciation ¹	(1,127)	(1,255)	(1,370)	(1,437)	(1,492)
Currency and unexplained changes ²	(695)	248	(14)	(12)	(297)
Property, plant, and equipment, end of year	15,401	17,043	18,161	19,681	20,890

¹ Reported in the statement of cash flows.² Calculated as the unexplained difference between beginning and end of year.

also can affect the change in accounts. For example, companies translate foreign balance sheets into their home currencies, so changes in accounts will capture both true investments (which involve cash) and currency-based restatements (which are merely accounting adjustments and not the flow of cash into or out of the company). If a particular account is a significant part of cash flow, use the cash flow statement and notes from the annual report to better understand the year-to-year change in the account.

Exhibit 11.14 deconstructs the change in property, plant, and equipment for Costco. Capital expenditures and asset dispositions are reported in the accountant's cash flow statement. To estimate depreciation, start with depreciation and amortization from the cash flow statement and, if amortization of acquired intangibles exists, subtract it (it is often found in the note on goodwill and intangible assets). The remaining line items are found in the management discussion and analysis, or when not disclosed, they have been estimated.

It is not always possible to eliminate the currency effects for each line item on the balance sheet. If this is the case, adjust aggregate free cash flow for currency effects using the balance sheet account titled foreign-currency translation, which under U.S. GAAP and IFRS is found within the statement of accumulated other comprehensive income. Unfortunately, the balance sheet account reports the aggregate effect across *all* foreign assets and liabilities, not just operating items. If you believe most currency adjustments are related to operating items, add the increase in the currency translation account to determine free cash flow. Consider the situation where inventory is rising on the balance sheet due to currency changes and not investment. To balance the balance sheet, the company increases the currency translation account within equity. Since the increase in inventory overstates actual investment in inventory, adding the increase in foreign-currency translation back to free cash flow undoes the negative cash flow caused by currency translation. For Costco, since we adjusted critical accounts one by one, we classify the unexplained currency translations as nonoperating.

Cash Flow Available to Investors

Although not included in free cash flow, cash flows related to nonoperating assets are valuable in their own right. They must be evaluated and valued separately and then added to free cash flow to give the total cash flow available to investors:

Present Value		Value of		Total Value
of Company's	+	Nonoperating	=	of
Free Cash Flow		Assets		Enterprise

To reconcile free cash flow with total cash flow available to investors, include the following nonoperating cash flows:

- *Nonoperating income and expenses.* Unless you can net the account against a change in a corresponding asset or liability (because it is noncash), include nonoperating income and expenses in total cash flow available to investors, not in free cash flow.
- *Nonoperating taxes.* Include nonoperating taxes in total cash flow available to investors. Nonoperating taxes include tax shields on nonoperating items and other nonoperating taxes disclosed in the tax reconciliation table.
- *Cash flow related to excess cash and marketable securities.* Subtract the increase (or add the decrease) in excess cash and marketable securities to compute total cash flow available to investors. If the company reports unrecognized gains and losses related to marketable securities in its statement of other comprehensive income, net the gain or loss against the change computed previously.
- *Cash flow from other nonoperating assets.* Repeat the process used for excess cash and marketable securities for other nonoperating assets. When possible, combine nonoperating gains and losses from a particular asset with changes in that nonoperating asset.

Reconciling Cash Flow Available to Investors

Cash flow available to investors should be identical to the company's total financing flow. By modeling cash flow to and from investors, you will catch mistakes otherwise missed. Financial flows include flows related to debt, debt equivalents, and equity:

- *Interest expenses.* Interest from both traditional debt and operating leases should be treated as a financing flow.
- *Debt issues and repayments.* The change in debt represents the net borrowing or repayment on all the company's interest-bearing debt, including

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short-term debt, long-term debt, and capitalized operating leases. All changes in debt should be included in the reconciliation of total funds invested, not in free cash flow.

- *Change in debt equivalents.* Since accrued pension liabilities and accrued postretirement medical liabilities are considered debt equivalents (see Chapter 23 for more on issues related to pensions and other postretirement benefits), their changes should be treated as a financing flow.¹¹
- *Dividends.* Dividends include all cash dividends on common and preferred shares. Dividends paid in stock have no cash effects and should be ignored.
- *Share issues and repurchases.* When new equity is issued or shares are repurchased, four accounts will be affected: common stock, additional paid-in capital, treasury shares, and retained earnings (for shares that are retired). Although different transactions will have varying effects on the individual accounts, only the aggregate matters, not how the individual accounts are affected. Exhibit 11.13 refers to the aggregate change as “Repurchases of common stock.”
- *Outflows to nonconsolidated subsidiaries.* Income attributable to nonconsolidated subsidiaries, found at the bottom of the income statement, is a financing flow, similar to dividends.

ADVANCED ISSUES

In this section, we summarize a set of the most common advanced topics in reorganizing a company's financial statements, including nonoperating charges and restructuring reserves, operating leases, pensions, and capitalized research and development (R&D). We provide only a brief summary of these topics here, as each one is discussed in depth in the chapters of Part Three, “Advanced Valuation Techniques.”

Nonoperating Charges and Restructuring Reserves Provisions are noncash expenses that reflect future costs or expected losses. Companies record provisions by reducing current income and setting up a corresponding reserve as a liability (or deducting the amount from the relevant asset).

For the purpose of analyzing and valuing a company, we categorize provisions into one of four types: ongoing operating provisions, long-term operating provisions, nonoperating restructuring provisions, and provisions created for the purpose of smoothing income (transferring income from one period to

¹¹ Pensions will affect many accounts, including the pension expense on the income statement, pension assets, pension liabilities, and deferred taxes. Exhibit 11.16, shown later in this chapter, aggregates each of the pension accounts into a single number for the cash flow statement.

another). Based on the characteristics of each provision, adjust the financial statements to reflect the company's true operating performance:

- *Ongoing operating provisions.* Operating provisions such as product warranties are part of operations. Therefore, deduct the provision from revenue to determine NOPAT, and deduct the corresponding reserve from net operating assets to determine invested capital.
- *Long-term operating provisions.* For certain liabilities, such as expected plant decommissioning costs, deduct the operating portion from revenue to determine NOPAT, and treat the interest portion as nonoperating. Treat the corresponding reserve as a debt equivalent.
- *Nonoperating provisions.* Unless deemed as ongoing, provisions such as one-time restructuring charges related to severance are nonoperating. Treat the expense as nonoperating and the corresponding reserve as a debt equivalent.
- *Income-smoothing provisions.* Classify any provisions identified for the purpose of income smoothing as nonoperating, and their corresponding reserve as an equity equivalent. Since income-smoothing provisions are noncash, they do not affect value.

The process for classifying and properly adjusting for provisions is explained in more detail in Chapter 21.

Operating Leases

Starting in 2019, companies that report under U.S. Generally Accepted Accounting Principles (GAAP) or International Financial Reporting Standards (IFRS) are required to capitalize nearly all asset leases, including short-term ones known as operating leases.¹² Under both standards, the present value of operating lease payments will be recorded on the balance sheet. On the income statement, IFRS allocates operating lease payments to depreciation and interest expense as appropriate, so no adjustment is necessary. For companies using GAAP, the entire lease expense, including embedded interest, is incorporated into other operating expenses like cost of sales. If this is the case, reclassify embedded interest in the lease expense as an interest expense.

Since past statements will not be restated, make sure to adjust them for operating leases to create a like-for-like analysis. To estimate the present value of operating leases prior to adoption of the new standard, search the notes for future rental commitments. Costco reports rental commitments in its note on

¹² The International Accounting Standards Board (IASB) published IFRS 16, "Leases," in January 2016, and the Financial Accounting Standards Board (FASB) issued Accounting Standards Update (ASU) 2016-02, "Leases" (Topic 842), in February 2016.

EXHIBIT 11.15 **Costco: Impact of Capitalizing Operating Leases on ROIC**

\$ million

	2015	2016	2017	2018	2019
EBITA					
EBITA, using rental expense	3,624	3,672	4,111	4,480	4,737
Implied interest expense ¹	73	75	57	74	91
EBITA, adjusted for operating leases	3,967	3,747	4,168	4,554	4,828
Yield-to-maturity on 10-year AA-rated debt	3.19%	3.36%	2.44%	2.91%	3.63%
Operating cash taxes					
Operating cash taxes, using rental expense	1,156	1,121	1,471	1,434	987
Tax shield on implied interest expense ²	27	28	21	21	22
Operating cash taxes, adjusted for operating leases	1,184	1,149	1,493	1,455	1,009
NOPAT					
NOPAT, using rental expense	2,468	2,551	2,640	3,046	3,750
After-tax implied interest expense	46	47	35	52	68
NOPAT, adjusted for operating leases	2,513	2,598	2,675	3,098	3,818
Invested capital					
Invested capital, without operating leases	13,023	15,607	14,978	15,651	16,583
Capitalized operating leases ³	2,230	2,320	2,528	2,500	2,414
Invested capital, including capitalized operating leases	15,253	17,928	17,506	18,151	18,997
ROIC, using beginning-of-year capital					
ROIC, using rental expenses	19.5%	19.6%	16.9%	20.3%	24.0%
ROIC, adjusted for operating leases	16.8%	17.0%	14.9%	17.7%	21.0%

¹ Implied interest is calculated by multiplying the yield-to-maturity of 10-year AA-rated debt by the beginning-of-year capitalized operating leases.

² The tax shield on implied interest expense is calculated by multiplying implied interest expense by the statutory tax rate. The statutory tax rate is reported in Exhibit 11.10.

³ Capitalized operating leases are estimated for 2019 in Exhibit 22.10.

leases. Discount each future rental commitment by an interest rate on low-risk debt to determine the present value of operating leases. Since companies report only five years of payments and aggregate the remaining payments into a single number, use an annuity to value remaining payments beyond the first year.

Exhibit 11.15 presents the adjustment for operating leases for Costco's historical statements.¹³ The present value of lease payments for Costco in 2018 equals \$2.5 billion. To determine interest embedded in 2019 EBITA, multiply 2018 capitalized operating leases by the rate of secured debt. (Given the ease of repossessing capital for operating leases, use an AA interest rate for

¹³ Because Costco's fiscal year ends prior to December 15, the company will not adopt the new leasing standard until 2020. Therefore, the value of operating leases must be estimated for historical years prior to 2020. For companies whose fiscal years end after December 15, no adjustment is required for 2019.

discounting and estimating embedded interest.) Next, adjust operating taxes to eliminate the tax shield related to implied interest. Subtract adjusted operating taxes from adjusted EBITA to determine NOPAT, adjusted for leases. Note how capitalizing operating leases increases both NOPAT and invested capital. The increase is not symmetric, causing ROIC to fall for Costco. This is because operating leases are a form of debt. For companies earning a return greater than their cost of debt, leverage artificially increases returns.

While capitalizing operating leases improves the quality of benchmarking, whether or not you capitalize will not affect intrinsic value as long as it is incorporated correctly in free cash flow, the cost of capital, and debt equivalents. Chapter 22 demonstrates how to incorporate operating leases throughout the valuation. The chapter also discusses alternative models to value operating leases.

Retirement Obligations Such as Pensions

Following the passage of FASB Statement 158 under U.S. GAAP in 2006, companies now report the present value of pension shortfalls (and excess pension assets) on their balance sheets.¹⁴ Since excess pension assets do not generate operating profits, nor do pension shortfalls fund operations, pension accounts should not be included in invested capital. Instead, pension assets should be treated as nonoperating assets, and pension shortfalls as a debt equivalent (and both should be valued separately from operations). If pension accounts are not explicitly detailed on the company's balance sheet, search the pension footnote to determine where they are embedded. Often excess pension assets are embedded in other assets, and unfunded pension liabilities are in other liabilities.

Reporting rules under IFRS (IAS 19) differ slightly in that companies can postpone recognition of their unfunded pension obligations resulting from changes in actuarial assumptions, but only as long as the cumulative unrecognized gain or loss does not exceed 10 percent of the obligations. For companies reporting under IFRS, search the notes for the current value of obligations.

On the income statement, new GAAP accounting for pensions in 2018 dictates that only service cost—the new benefits promised to employees for service rendered in a given year—be included in operating expenses like cost of goods sold.¹⁵ The remaining items, such as expected return on assets and interest cost on the liabilities, are now included as nonoperating income or expense. For years prior to 2018, an adjustment is still required.

¹⁴ From December 2006, FASB Statement 158 eliminated pension smoothing on the balance sheet. Companies are now required to report excess pension assets and unfunded pension obligations on the balance sheet at their current values, not as smoothed values as in the past.

¹⁵ The FASB published ASU 2017-07, "Compensation—Retirement Benefits (Topic 715): Improving the Presentation of Net Periodic Pension Cost and Net Periodic Postretirement Benefit Cost," on March 10, 2017. IFRS already separates service cost from financial performance in pensions.

Since Costco does not provide pension benefits to employees, we do not adjust the company's historical statements. Chapter 23 provides details on how to adjust NOPAT for pensions and how to factor under- or overfunded pensions into a company's value.

Capitalized Research and Development

In line with the conservative principles of accounting, accountants expense R&D, advertising, and certain other costs in their entirety in the period when they are incurred, even when economic benefits resulting from such expenses continue beyond the current reporting period.¹⁶ This practice can dramatically understate invested capital and overstate return on capital for some companies. Therefore, you should consider whether it would be effective to capitalize and amortize R&D and other quasi investments in a manner like that used for capital expenditures. Equity should be adjusted correspondingly to balance the invested-capital equation.

If you decide to capitalize R&D, do *not* deduct the reported R&D expense from revenue to calculate operating profit. Instead, deduct the amortization associated with past R&D investments, using a reasonable amortization schedule. Since amortization is based on past investments (versus expense, which is based on current outlays), this approach will prevent reductions in R&D from driving short-term improvements in ROIC.

Whether or not you capitalize certain expenses will not affect computed value; it will affect only the timing of ROIC and economic profit. Chapter 24 analyzes the complete valuation process for R&D-intensive companies, including adjustments to free cash flow and value.

Other Advanced Adjustments

Some companies may have industry-specific items that require adjustment. These adjustments arise from an uncommon line item on the income statement or balance sheet and, given their rarity, require thoughtful judgment based on the economic principles of this book.

Consider an example from FedEx. In 2013, the company sold aircraft to another company and leased the aircraft back. This transaction is commonly known as a sale-leaseback. If a gain arises from the sale, the company cannot recognize the gain as income, but instead must lower the annual rental expense over the life of the contract. Since cash increases but retained earnings do not rise, a liability for deferred gains is recognized.

¹⁶ One exception to this conservatism is the development of software. Although software is an intangible asset, both GAAP and IFRS accounting allow for certain software investments to be capitalized and amortized over the life of the asset.

Should the liability for deferred gains be treated as operating and deducted from operating assets to determine invested capital? Or perhaps classified as a debt or equity equivalent? From a valuation perspective, it doesn't matter how to classify the item, as long as it is treated consistently. It will, however, have an impact on our perceptions about return on invested capital and ultimately value creation. Accounting rules prevent the one-year spike in income caused by a financial transaction, but we believe the downward distortion in future rental expense is worse, since this lower rental expense is noncash and could distort the perceived cost of new leases. Therefore, undo the transaction entirely and recognize the account as an equity equivalent.

Not every advanced issue will lead to material differences in ROIC, growth, and free cash flow. Before collecting extra data and estimating required unknowns, decide whether the adjustment will further your understanding of a company and its industry. An unnecessarily complex model can sometimes obscure the underlying economics that would be obvious in a simple model. Remember, the goal of financial analysis is to provide a strong context for good financial decision making and robust forecasting, not to create an overly engineered model that deftly handles unimportant adjustments.

REVIEW QUESTIONS

- Exhibit 11.16 presents the income statement and balance sheet for Companies A, B, and C. Compute each company's return on assets, return on equity, and return on invested capital. Based on the three ratios, which company has the best operating performance?

EXHIBIT 11.16 **Ratio Analysis: Consolidated Financial Statements**

\$ million

	Company A	Company B	Company C
Income statement			
Operating profit	100	100	100
Interest	0	0	(20)
Earnings before taxes	100	100	80
Taxes	(25)	(25)	(20)
Net income	75	75	60
Balance sheet			
Inventory	125	125	125
Property, plant, and equipment	400	400	400
Equity investments (cost method)	0	50	0
Total assets	525	575	525
Accounts payable	50	50	50
Debt	0	0	200
Equity	475	525	275
Liabilities and equity	525	575	525

- Why does the return on assets differ between Company A and Company B? Why do companies with equity investments tend to have a lower return on assets than companies with only core operations?
- Why does the return on equity differ between Company A and Company C? Is this difference attributable to operating performance? Which better reflects operating performance, return on assets or return on equity? Why?
- Exhibit 11.17 presents the income statement and balance sheet for HealthCo, a \$600 million health-care company. Compute NOPAT, average invested capital, and ROIC for the current year and next year. Assume an operating tax rate of 30 percent. If the weighted average cost of capital is 10 percent, is the company creating value? In which year is the firm performing better? Why?

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EXHIBIT 11.17 **HealthCo: Income Statement and Balance Sheet**

\$ million								
Income statement	Last year	Current year	Next year	Balance sheet	Last year	Current year	Next year	
Revenues	1,100	1,210	1,307	Operating cash	22	24	26	
Cost of sales	(770)	(871)	(915)	Excess cash and marketable securities	91	74	83	
Selling, general, and administrative	(165)	(182)	(196)	Accounts receivable	220	242	261	
Depreciation	(33)	(36)	(39)	Inventory	330	363	392	
EBIT	132	121	157	Current assets	663	703	763	
Interest expense	(15)	(15)	(15)	Property, plant, and equipment	440	484	523	
Gain/(loss) on sale of assets	-	(10)	-	Equity investments	50	50	50	
EBT	117	96	142					
Taxes	(35)	(29)	(43)					
Net income	82	67	99	Total assets	1,153	1,237	1,336	
Dividends	33	27	40	Accounts payable	275	303	327	
				Short-term debt	90	90	90	
				Accrued expenses	165	182	196	
				Current liabilities	530	574	613	
				Long-term debt	210	210	210	
				Common stock	100	100	100	
				Retained earnings	313	353	413	
				Total liabilities and equity	1,153	1,237	1,336	

- Using the reorganized financial statements created in Question 4, what is the free cash flow for HealthCo in the current year and next year?
- Many companies hold significant amounts of excess cash, or cash above the amount required for day-to-day operations. What would happen to the ROIC for HealthCo if you included excess cash in its calculation? Why should one exclude excess cash from the calculation?